
Exercice 01

Mar 3, 2026 • Advanced Database

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Database Researcher

Speaker

SQL Query

```
SELECT DISTINCT
  p.PropertyNum,
  p.City,
  p.Type,
  p.Rental_Price
FROM Property p
JOIN Visit v
  ON v.PropertyNum = p.PropertyNum
JOIN Customer c
  ON c.CustomerNum = v.CustomerNum
WHERE p.OwnerNum = 'CP93'
  AND p.Type = 'Apartment'
  AND p.Type = c.PrefType
  AND p.Rental_Price <= c.BudgetMax;
```

b. Algebraic tree

Query Tree

```

      π {p.PropertyNum, p.City, p.Type, p.Rental_Price}
      |
      σ {o.OwnerNum = 'CP93' ∧ p.Type = 'Apartment'
        ∧ p.Type = c.PrefType ∧ p.Rental_Price ≤ c.BudgetMax}
      |
      ⋈ {v.CustomerNum = c.CustomerNum}
      /      \
      ⋈ {p.PropertyNum = v.PropertyNum}      Customer (c)
      /      \
      ⋈ {o.OwnerNum = p.OwnerNum}      Visit (v)
      /      \
      Owner (o)      Property (p)
```

Observation

- 1 Selection sits above all joins
- 2 Large intermediate results are produced before filtering

Transformation rules

1 Selection Pushdown Rule

2 Splitting Selections

3 Reorder joins

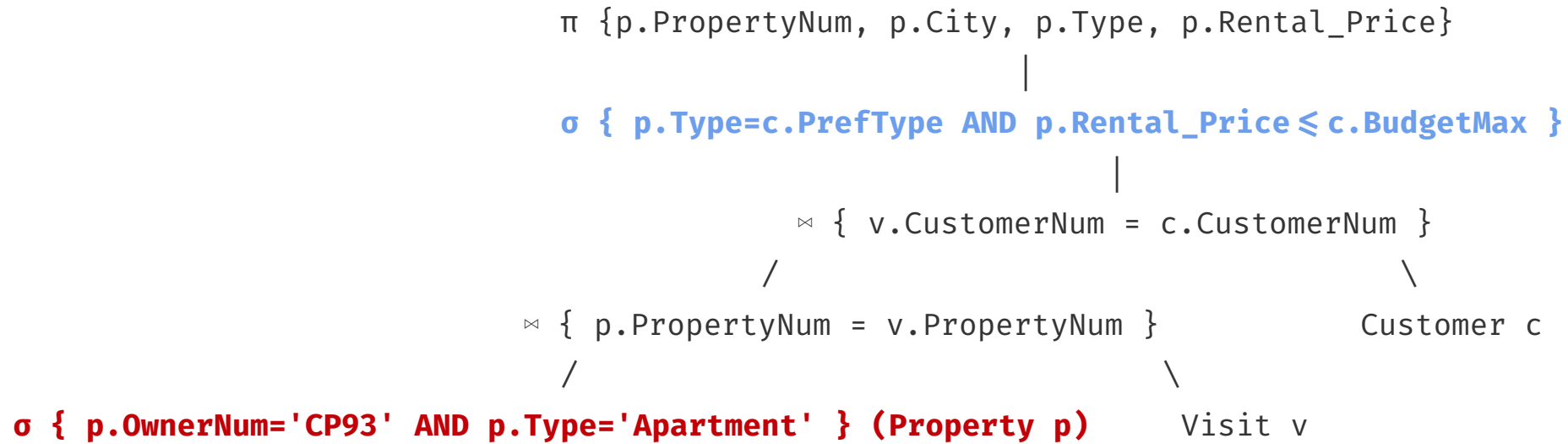


Core Idea

Reduce intermediate tuples early

Optimized algebraic Tree

Query Tree



Exercice 02

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Cost Calculation Methodology

A systematic three-step approach to evaluating query execution plans through distinct cost metrics.

1

I/O Cost

Number of pages loaded from disk to execute the operation

2

Tuple Cost

Number of tuples generated by the operation

3

Storage Cost

Number of pages needed to store the tuples generated by the operation (based on the number of tuples from step 2)

Important Note

Storage costs cannot be computed without tuple size and page size information. Therefore, some intermediate storage costs remain non-computable in this analysis.

tuple cost=100rows

storage cost????????number of page

page size=10octets

$R(a,b,c)$ $\text{size}(a)+\text{size}(b)+\text{size}(c)=15\text{octets}$

====>tuple size

result size= $15*100=1500$ octets

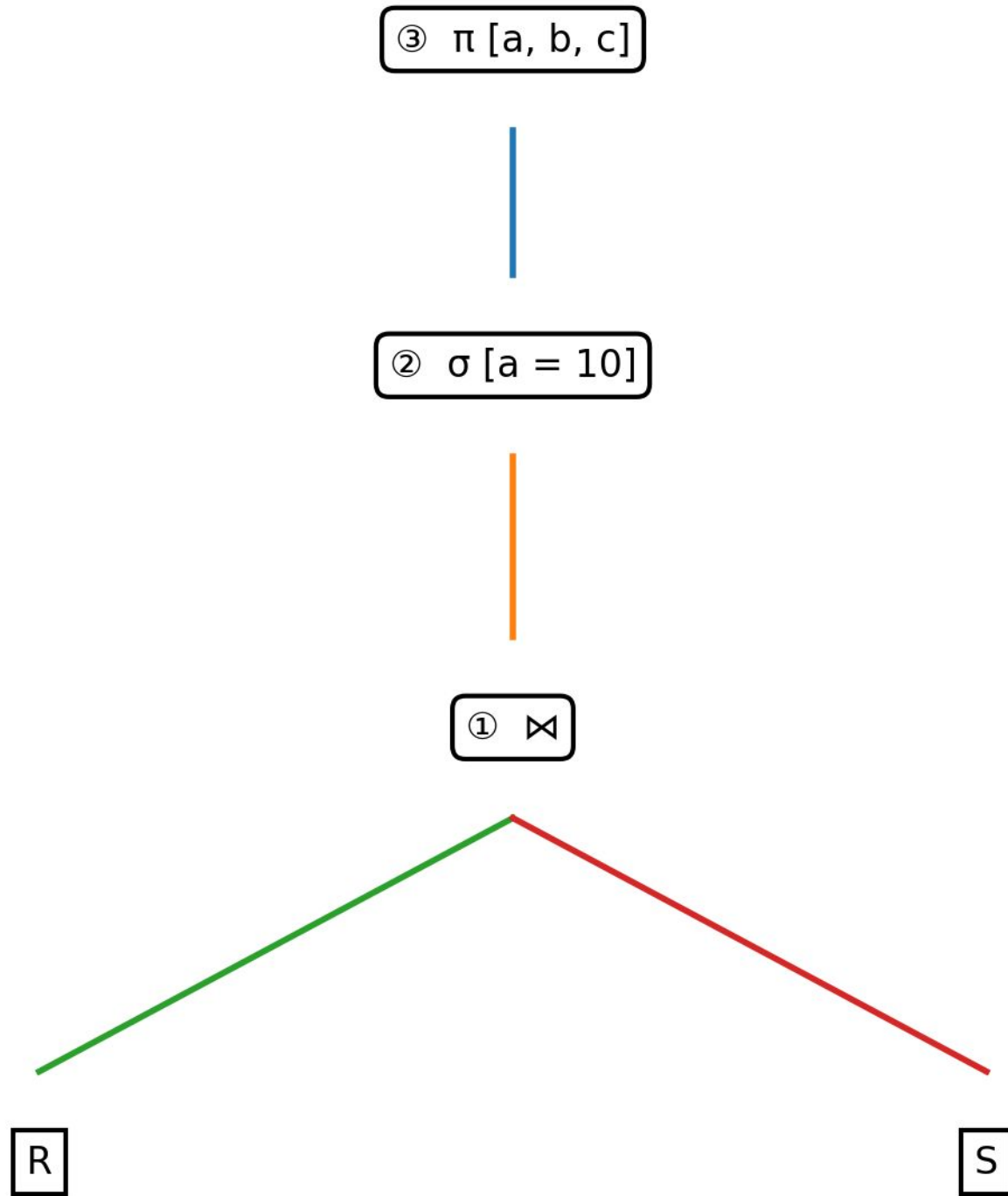
storage cost= $1500/10=150$ pages

informations:

Assumption: 10 rows = 1 I/O (1 page)

Relation	Rows	Val
R	5,000 rows	Val(R,a)=50 val(R,b)=100
S	2000 rows	Val(S,b)=200 val(S,c)=100

** Pages are calculated as Rows / 10*



Plan A — Step-by-step I/O Cost Table

Plan	Step	Operation	I/O operations	I/O cost	Resulting set rows	Total I/O cost
A	A1	Join R⋈S				
A	A2	Select a=10 (scan join result)				
A	A3	Project πa,b,c (scan result)				

Plan A Total Cost:

Tuple Cost

number of tuples generated by the operation

General Formula

$$|R \bowtie S| = |R| * |S| / \max(\text{val}(R, b), \text{val}(S, b))$$

Calculation

$|R| = 5000$

$|S| = 2000$



$$5000 * 2000 / 200 =$$

Total Cost

50 000 rows

Outer Loop: R (500 pages)

×

Inner Loop: S (200 pages)

I/O cost

General Formula

Assumption: Join Cost Model (Page Nested-Loop). 10 tuples/block

$$\text{Cost}(R \bowtie S) = B(R) + B(R) \cdot B(S)$$

$$B(R) = 500$$

$$B(S) = 200$$



Calculation

$$500 + 500 \cdot 200 =$$

Total Cost

100,500 I/O

Outer Loop: R (500 pages)

×

Inner Loop: S (200 pages)

Plan A — Step-by-step I/O Cost Table

Plan	Step	Operation	I/O operations	I/O cost	Resulting set rows	Total I/O cost
A	A1	Join $R \bowtie S$	$B(R)+B(R) \cdot B(S)=500+500 \cdot 200$	100,500	50,000	100,500
A	A2	Select $a=10$ (scan join result)	Scan $(R \bowtie S)=B(R \bowtie S)=5,000$	5,000	1,000	105,500
A	A3	Project $\pi_{a,b,c}$ (scan result)	Scan result $B=100$	100	1,000	105,600

Plan A Total Cost: 105,600 I/O

condition $a=10$

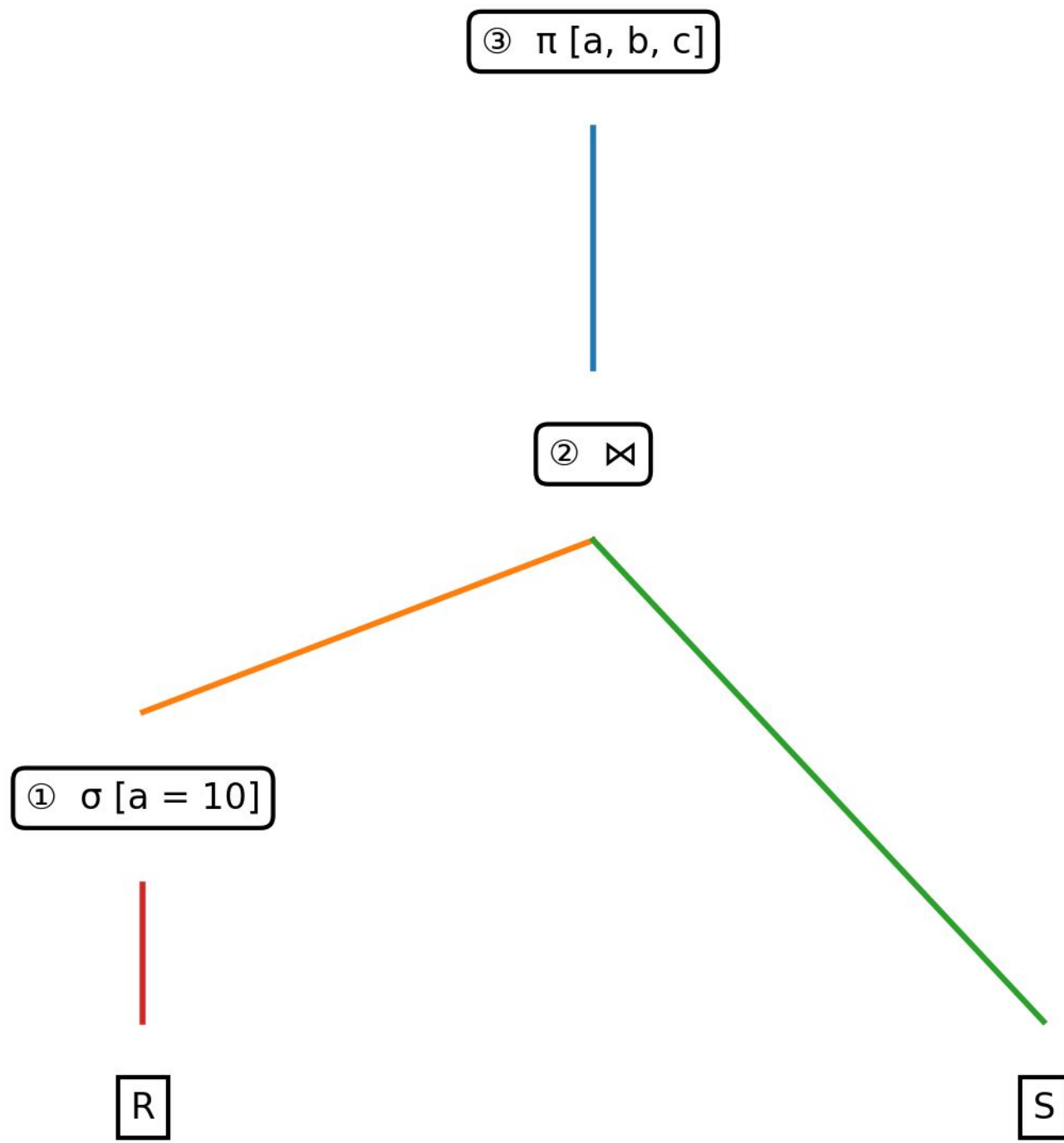
$\text{cost}(\text{select}(R)) = |R| * 1/\text{val}(R,a) = 500/50$

$SF = 1/\text{val}(R,a)$

$a=b$

And

OR



Plan B — Step-by-step I/O Cost Table

Plan	Step	Operation	I/O operations	I/O cost	Resulting set rows	Total I/O cost
B	B1	Select rows in R with a=10	Scan R = B(R)=500	500	100	500
B	B2	Join $(\sigma_R) \bowtie S$	$B(\sigma_R) + B(\sigma_R) \cdot B(S) = 10 + 10 \cdot 200$	2,010	1,000	2,510
B	B3	Project $\pi_{a,b,c}$ (scan result)	Scan result B=100	100	1,000	2,610

Plan B Total Cost: 2,610 I/O



Exercice 03

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SQL Query

```
SELECT s.name FROM Student s, Book b, Borrow br WHERE  
AND b.author = 'DATE' AND s.age = 20
```

Tables

 Student	10,000 tuples 1,000 pages · 10 tuples/page
 Book	50,000 tuples 5,000 pages · 10 tuples/page
 Borrow	300,000 tuples 15,000 pages · 20 tuples/page

Selectivity Factors

500 different authors
author='DATE'

1/500

ages 10–24 (15 values)
age=20

1/15

Cost Calculation Methodology

A systematic three-step approach to evaluating query execution plans through distinct cost metrics.

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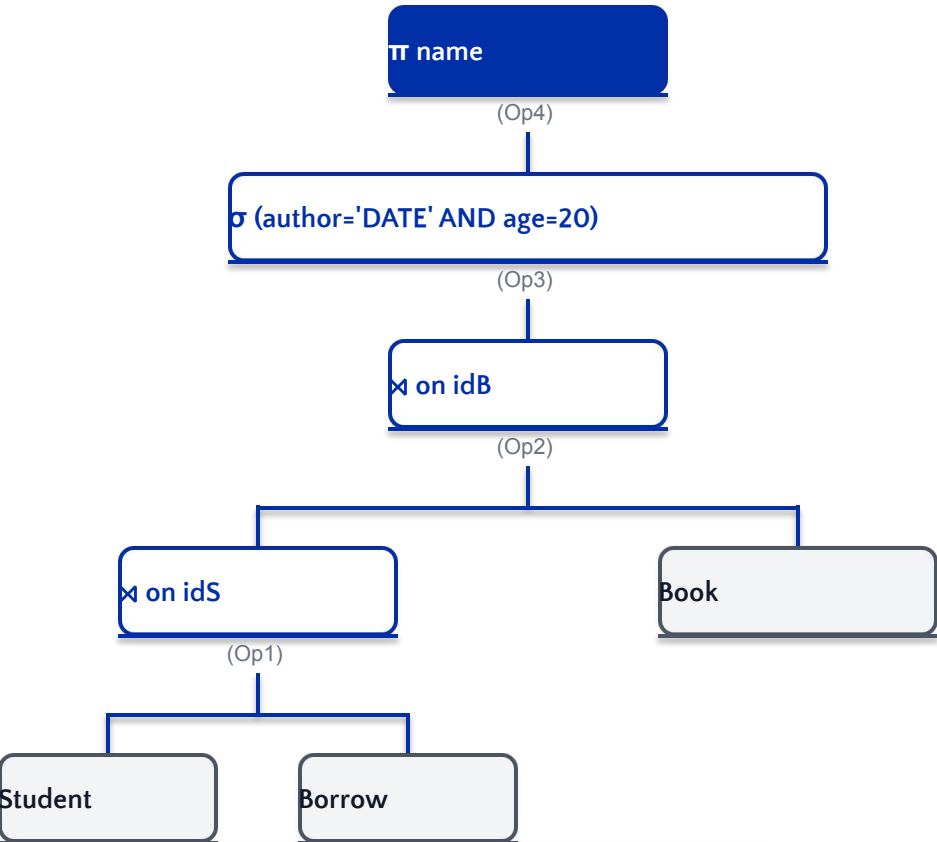
Storage Cost

Number of pages needed to store the tuples generated by the operation (based on the number of tuples from step 2)

Important Note

Storage costs cannot be computed without tuple size and page size information. Therefore, some intermediate storage costs remain non-computable in this analysis.

PLAN 1 - Execution Tree (No Indexes)



Op1: Join Student with Borrow on idS

Op2: Join result with Book on idB

Op3: Apply selection (author='DATE' AND age=20)

Op4: Project name

Note: Joins are executed before selections. No indexes available on join or selection attributes.

Operation 1 - Join Student with Borrow



Join Method: Nested Loop Join

I/O Cost Formula:

$$|R| + |R| \times |S|$$

Hash join $3 \times (|R| + |S|)$

Applied:

$$|Student| + |Student| \times |Borrow| = 1,000 + 1,000 \times 15,000$$

= 15,001,000 page I/Os



Tuple Cardinality Formula:

$$|Op1| = (|Student| \times |Borrow|) / \max(\text{val}(Student, idS), \text{val}(Borrow, idS))$$

Calculation:

$$(10,000 \times 300,000) / 10,000$$

= 300,000 tuples



Storage Cost:

Not computable - Tuple size after join is unknown and page size is not provided

Operation 2 - Join with Book



1) Cost in number of I/O (pages loaded)

Case 1: RAM Not Saturated

Op1 pages already in RAM, only load Book

$$\text{nbTuples(Op1)} \times |\text{Book}|$$

$$30,000 \times 5,000 = 1,500,000,000 \text{ I/O operations}$$

Case 2: RAM Saturated

$$|\text{Op1}| + |\text{Op1}| \times |\text{Book}|$$



2) Cost in number of tuples generated

$$|\text{Op2}| = (|\text{Op1}| \times |\text{Book}|) / \max(\text{val}(\text{Op1}, \text{idB}), \text{val}(\text{Book}, \text{idB}))$$

$$(300,000 \times 50,000) / 50,000 = 300,000 \text{ tuples}$$



3) Number of pages to store tuples of Op2:

✗ Not computable

3 Operation 3: Selection

Condition

author='DATE' AND age=20

I/O Cost

0 (RAM not saturated)

Tuples Generated

Selectivity: author = $1/500$, age = $1/15$
 $(1/500) \times (1/15) \times 300,000$

= 40 tuples

Note: Ages range 10-24 (15 values)

4 Operation 4: Projection

Attribute

$\pi(\text{name})$

I/O Cost

0 (RAM not saturated)

Tuples Generated

$|\text{Op4}| = |\text{Op3}|$

= 40 tuples

Cost in bytes:

$40 \times \text{size}(\text{name})$
(if name size given)



Final Result Size

After all operations in Plan 1

40 tuples

Plan 1 Summary (No Indexes)

Operation	I/O Cost (Pages)	Tuples	Storage (Pages)
Op1	15,001,000	300,000	Not computable
Op2	1,500,000,000 (RAM not saturated)	300,000	Not computable
Op3	0 (Selection)	40	Not computable
Op4	0 (Projection)	40	Depends on name size

Op1: Student ⋈ Borrow

Nested loop join on idS

Op2: Op1 ⋈ Book

Join on idB (RAM not saturated)

Op3: Selection

author='DATE' AND age=20

Op4: Projection

Project name attribute

Total I/O Cost

1,515,001,000

page I/Os



Extremely High Cost

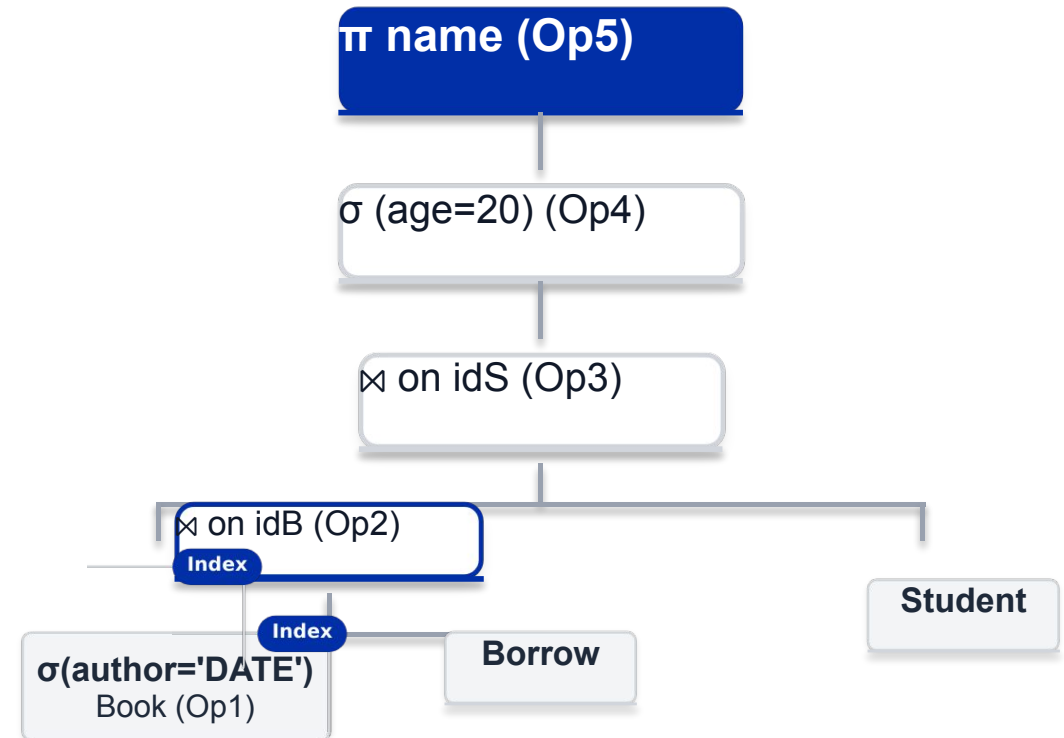
PLAN 2 - EXECUTION TREE (WITH INDEXES)

Indexes Used

- 🔑 Index on Book(author)
- 🔑 Index on Borrow(idB)

Execution Steps

- Op1:** Filter Book using index on author
- Op2:** filtered Book with Borrow using index on idB
- Op3:** Join result with Student on idS
- Op4:** Apply selection on age=20
- Op5:** Project name attribute



Operations 1 and 2 - Indexed Selection and Join

♦ Op1: $\sigma(\text{author}='DATE')$ using index on Book(author)

1) Cost in I/O

TD formula (index in RAM):

$|\text{Book}| \times \text{selectivity}(\text{author}) = 50,000 \times (1/500) = 100$ tuples, if we consider that each tuple is in on page we get **100 pages**

2) Tuples

✓ Op1 tuples = **100**

♦ Op2: Join (filtered Book) $\bowtie$ Borrow using index on Borrow(idB)

1) Cost in I/O

TD computes Borrow tuples per one Book tuple:

$300,000 \times (1/50,000) = 6$ tuples === if we consider that each tuple is in on page we get **6 pages**

For 100 book tuples: $100 \times 1 =$ **100 I/Os**

2) Tuples

$|\text{Op2}| = (100 \times 300,000) / 50,000 =$ **600 tuples**

Operations 3–5: Join, Selection, Projection

Following the indexed access, we join with the Student table and apply the final selection and projection, resulting in a highly efficient execution plan.

STEP 3

Operation 3: Join Op2 with Student

I/O Cost

1,000 pages

Tuples

600

RAM not saturated → load only Student

Formula: $(10,000 \times 600) / 10,000 = 600$

STEP 4

Operation 4: Selection (age=20)

I/O Cost

0 pages

Tuples

40

RAM not saturated → 0 I/O

Formula: $600 \times (1/15) = 40$ (selectivity: age range 10-24 → 15 values)

STEP 5

Operation 5: Projection (name)

I/O Cost

0 pages

Final Tuples

40

RAM not saturated → 0 I/O

Final result: $|Op5| = |Op4| = 40$ tuples

PLAN 2 SUMMARY (WITH INDEXES)

Operation	Description	I/O Cost	Tuples
Op1	$\sigma(\text{author}=\text{'DATE'})$ on Book using index	100 I/Os	100 tuples (50,000 $\times$ 1/500)
Op2	Join filtered Book $\bowtie$ Borrow using index on Borrow(idB)	100 I/Os	600 tuples (1 page per book)
Op3	Join Op2 $\bowtie$ Student (RAM not saturated)	1,000 I/Os	600 tuples (load Student)
Op4	Selection on age=20 (updated from TD range)	0 I/Os	40 tuples (600 $\times$ 1/15)
Op5	Projection $\pi(\text{name})$ (final result)	0 I/Os	40 tuples

Total I/O Cost

1,200 page I/Os

Indexes used: Book(author) + Borrow(idB)
Result: Same final output (40 tuples) with dramatically reduced I/O cost

Plan 1 vs Plan 2: Detailed Comparison

Plan	Operation	I/O Cost	#Tuples	Notes
Plan 1 (No Indexes)	Op1: Student ⋈ Borrow	15,001,000	300,000	Nested loop join
	Op2: Op1 ⋈ Book	1,500,000,000	300,000	RAM not saturated
	Op3: Selection (author, age)	0	40	Selectivity: 1/500 × 1/15
	Op4: Projection (name)	0	40	Final result
	TOTAL	1,515,001,000	40	page I/Os
Plan 2 (With Indexes)	Op1: σ(author) on Book	100	100	Index on author
	Op2: Book ⋈ Borrow	100	600	Index on idB
	Op3: Op2 ⋈ Student	1,000	600	Load Student
	Op4: Selection (age=20)	0	40	Selectivity: 1/15
	Op5: Projection (name)	0	40	Final result
	TOTAL	1,200	40	page I/Os

💡 Conclusion

Both plans produce the same final output (40 tuples), but indexes reduce I/O cost by several orders of magnitude — from over 1.5 billion to just 1,200 page I/Os.

Conclusion & Key Takeaways

Summary of the cost analysis and optimization strategies explored in Exercise 03.



Index usage drastically improves query performance



Early selection and indexed joins reduce intermediate result sizes



Cost-based optimization is essential for efficient query execution